

Ngoc Thien Kim Nguyen

AI Engineer

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Summary

Final-year ECE student researching applied LLMs at the intersection of AI and hardware. First author of **REFINE-VerilogV2** — dual-SLM + LangGraph + EDA-in-the-loop — reaching **100% syntax / 66.9% functional pass@5** on RTLLM v1.1. Three peer-reviewed publications on LLM-assisted Verilog generation.

Featured Research

REFINE-VerilogV2 — Dual-SLM Framework for Verilog Generation, First-author research, graduation thesis 2025 – Present

- Fine-tuned **Qwen2.5-Coder-7B** with LoRA (rank=2048, bfloat16, 4096-ctx) into two role-separated adapters — **Generator SLM + Debugger SLM** — on dual NVIDIA RTX 5880 Ada GPUs using Hugging Face PEFT and Transformers.
- Built a **LangGraph** runtime (Generator → Sanitizer → Correcter) that routes Verilator/Icarus/UVM diagnostics back to the Debugger via Topmost Exception Debugging and an Exception-Debugging Trial Manager (EDTM).
- Curated the **690K-module Pyranet** corpus using Yosys-based capacity-aware filtering (cell count + cell-type diversity), removing manual per-tier loss weighting.
- Built an end-to-end **evaluation pipeline** across VerilogEval V2 (156 modules) and RTLLM v1.1 (29 designs), tracking syntax/functional pass@k, EDA toolchain success, per-class failure modes, and TED iteration cost.
- **Achieved 100% syntax pass@5 and 66.9% functional pass@5** on RTLLM v1.1; resolved 18/18 initial compile faults across 29 designs (+6.9 pts over prior REFINE-Verilog).
- **Stack:** Python, PyTorch, Hugging Face PEFT/Transformers, LoRA, LangGraph, Verilator, Icarus Verilog, Yosys, UVM.

Publications

N.-T.-K. Nguyen, V.-M.-T. Nguyen, C.-K. Pham, D.-H. Le. “REFINE-VerilogV2: Enhancement in Partial Verilog Generation Leveraging Small Language Model in Combination with EDA Debugging.” *Under review*, 2026. (first author)

V.-M.-T. Nguyen, **N.-T.-K. Nguyen**, D.-H. Le. “REFINE-Verilog: Reliability-based fine-tuning and inference enhancement for SLM in partial Verilog generation.” *Int. Conf. on Information and Communication Technology (ICICT)*, 2026.

V.-M.-T. Nguyen, **N.-T.-K. Nguyen**, D.-H. Le. “COMBA-PROMPT: Comprehensive benchmark and augmentation for Verilog generation leveraging dual-LLM prompting.” *Int. Conf. on Multimedia, Computing and Technology for Sustainable Development (MCT4SD)*, 2025.

Research Experience

Undergraduate Researcher, DESLab Laboratory, University of Science (VNUHCM) Apr 2023 – Present

- Built a quadruped robot end-to-end: 3D-printed hardware, motor control (STM32/ESP32), and PCB design for motor drivers.
- Designed and laid out analog ICs (Inverter, NAND, NOR, XOR, SRAM, Full Adder) with pre/post-layout simulation in Cadence Virtuoso.

Skills

AI / ML: LLM & SLM Fine-tuning, LoRA/QLoRA, LangGraph, Agents, RAG, Prompt Engineering, PyTorch, Hugging Face

Programming: Python, Pandas, Embedded C

Systems / Hardware: STM32, ESP32, Arduino, PCB Design (Altium, KiCad), Cadence Virtuoso

Education

B.Eng. Electronics & Telecommunications — VNUHCM University of Science Sep 2021 – Sep 2026

JUSST Exchange — University of Electro-Communications, Japan Apr 2025 – Sep 2025

Languages

Vietnamese (native) • English (professional working)

Certifications

AI Engineer Core Track — LLM Engineering, RAG, QLoRA, Agents (2025)